

SECTION - A

1. (b) 8
2. (b) Contains all elements under consideration
3. (a) $4/5$
4. (c) Many - One relation
5. (b) $1/2$
6. (a) 10
7. (b) $A \cap B = \emptyset$
8. (a) Reflexive.

SECTION - B - Very Short Answers.

9. A set is collection of objects written in braces
10. Domain of $f(x) = x^2$ is real numbers and range is real numbers.
11. $\sin 45^\circ \cos 45^\circ = 1/2$
12. $5p^2 = 20$.
13. Number of elements in power set is 16.
14. One - one function is $f(x) = x$ and many - one is $f(x) = x^2$

SECTION - C - Short Answers

15. LHS = $\sin\theta/(1+\cos\theta) + (1+\cos\theta)/\sin\theta$
After simplification,
RHS = $2 \operatorname{cosec}\theta$.

16. $A \cup B = \{1, 2, 3, 4, 5, 6\}$ and $A \cap B = \{3, 4\}$.

17. Total numbers = $5P3 = 60$,

18. Relation is reflexive and symmetric.

SECTION - D - Long Answer.

19. $(1+\tan^2\theta)(1+\sin^2\theta)$

Using, $1+\tan^2\theta = \sec^2\theta$
LHS = RHS,